Mela [መላ]

"Your Stipend, Your Plan, Your Peace of Mind: Navigate University Life with Financial Clarity."

Faculty of Informatics

IT- 3rd year | Mobile Application Design Project Proposal

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Problem of the statement: What's the problem?

In Ethiopia, public university education is a transformative opportunity for thousands of students from diverse socioeconomic backgrounds. One of these opportunities is the government provided monthly stipend; a financial lifeline designed to cover the basic living expenses of students throughout the academic year. For the majority of students, this stipend is not merely supplementary income; it is their only source of financial support. It must cover everything from daily meals and academic materials to personal care and local transportation. Any disruption or miscalculation in managing these funds can have immediate and severe consequences on a student's well-being and academic performance.

However, the economic situation surrounding Ethiopian university campuses has shifted dramatically in recent years. The cost of basic necessities, the essential goods and services a student requires to simply exist and study has experienced relentless and significant inflation. What was once a manageable stipend has now become a source of constant anxiety for students trying to stretch increasingly thin resources across an entire month.

Consider the specific financial pressures students face daily:

- **Daily Food Expenses:** The price of meals at university cafeterias and local eateries has risen dramatically. A student who could previously afford three meals a day now finds themselves carefully rationing their food budget, sometimes skipping meals to make ends meet. The cost of staple items like injera, bread, and cooking oil continues to climb, directly impacting a student's ability to maintain proper nutrition.
- **Academic Materials:** Education requires more than just tuition. Students must purchase notebooks, pens, printed lecture notes, photocopies of reference materials, and stationery for assignments and projects. These costs are non-

negotiable—falling behind on academic materials directly affects a student's ability to participate in coursework and perform well in examinations. Yet, the prices of these essential academic supplies have risen sharply, forcing students to choose between buying a textbook and having enough for transportation home.

- **Essential Personal Care:** Hygiene is not a luxury; it is a necessity. Soap, shampoo, sanitary products for female students, laundry detergent, and other personal care items are essential for health, dignity, and social well-being. However, these items have become increasingly expensive, and students on tight budgets often find themselves postponing purchases or using substandard alternatives to save money.
- **Local Transportation:** University campuses are often located on the outskirts of towns, requiring students to travel into the city for medical appointments, banking, purchasing supplies, or seeking part-time work opportunities. The cost of local transport whether by bus, minibus, or *Bajaj* adds up quickly. A single round trip can consume a significant portion of a day's food budget, leading students to avoid essential errands or limit their mobility.

Beyond these individual expenses, there is a deeper psychological toll. Students live in a state of chronic financial precarity. They must constantly perform mental calculations: *If I buy this notebook today, will I have enough for lunch for the rest of the week? If I take this bajaj to the clinic, what do I sacrifice tomorrow?* This constant mental load often referred to as "financial scarcity mindset", which consumes cognitive bandwidth that should be dedicated to learning, problem-solving, and academic growth.

The consequences are tangible and widespread:

- Students frequently run out of money before the end of the month, forcing them to borrow from friends, skip meals, or fall behind on essential purchases.
- Financial stress contributes to anxiety, depression, and a diminished capacity to focus on studies.
- Students are unable to build any form of savings, leaving them vulnerable to unexpected expenses like illness or urgent academic fees.

Currently, students have no tool designed specifically for their unique financial reality. They rely on mental math, scattered notes in notebooks, or generic budgeting apps that do not account for the fixed stipend structure, the rising cost of local necessities, or the cultural and economic context of Ethiopian university life. These makeshift solutions are insufficient and often fail precisely when they are needed most.

Mela is designed to fill this critical gap. By providing a tool that understands the constraints of the student stipend, prioritizes essential spending categories, and offers proactive, non-judgmental guidance, we aim to transform financial management from a source of stress into a source of empowerment. Our goal is to help students like Biruk move from a reactive survival mode to a proactive, planned approach, freeing their minds to focus on what truly matters: their education and their future.

3. BRIEF USER PERSONA: What makes us decide to do it?

Our application serves Ethiopian university students who struggle to manage their fixed stipend against rising living costs. To ensure our solution addresses real needs, we have developed two primary personas representing our target users.

Persona 1: Biruk – The Overwhelmed Upper-Year Student

Biruk is a 22-year-old 4th-year Electrical Engineering student at Hawassa University. He receives a monthly stipend of 4,000 Birr, which must cover all his expenses like food, printing, transportation, and personal care. In his final year, he is juggling a demanding capstone project, heavy coursework, and the stress of job hunting. Over the past two years, he has watched his purchasing power shrink as the cost of basic necessities has steadily risen. A meal that cost him 70 Birr last year now costs 100 Birr. Printing lecture notes and circuit diagrams for his project has become a recurring expense that eats into his food budget. Transportation for electronic components costs him 60 Birr round trip money he often has to take from his daily food allowance. He frequently

runs out of money before month end, forcing him to skip meals or borrow from friends, which adds social stress to his financial worries. He needs a tool that helps him track his spending, warns him before he overspends, and allows him to save even a small amount for his graduation expenses.

Persona 2: Hiwot – The Inexperienced First-Year Student

Hiwot is a 19-year-old Freshman student at Hawassa University. She comes from a small town in the Gurage Zone and is managing her own finances for the first time in her life. Her monthly stipend is 3,000 Birr, and she receives no additional support from her family. Living away from home has been both liberating and overwhelming. She often underestimates how quickly small purchases add up a coffee here, a snack there, a notebook she forgot to buy, and finds herself struggling by the third week of the month. As a young woman, she has essential personal care costs, such as sanitary products, which can cost her 150-200 Birr monthly—a significant portion of her stipend that her male peers do not have to account for. She also faces social pressure to participate in group outings with new friends, but she often declines because she cannot afford it. She needs a simple, guided tool to help her understand where her money goes, prioritize her essentials, and build confidence in managing her finances independently so she can focus on her studies without constant anxiety.

These two personas represent the range of students we aim to serve—from the experienced but overwhelmed upper-year student facing the compounded pressure of inflation and final-year demands, to the first-year student navigating financial independence for the first time while managing gender-specific expenses. Both share the common struggle of making a fixed stipend last in an environment where the cost of basic necessities continues to rise.

4. **PROPOSED SOLUTION:** What's better?

Mela is a dedicated financial scheduler built to address the specific economic reality of Ethiopian university students. It provides clarity, control, and peace of mind in an environment of fixed income and rising costs. The app moves beyond simple transaction tracking to offer proactive, intelligent planning tailored to a student's unique financial lifecycle.

Unlike generic budgeting apps that assume variable income and cater to salaried professionals, Mela is designed from the ground up with the Ethiopian student experience in mind. It understands that the stipend arrives as a single lump sum, that expenses fall into predictable yet essential categories, and that the rising cost of basic necessities requires constant vigilance. The app serves as a financial companion—non-judgmental, supportive, and always available to help students make informed decisions.

The solution is built on five core principles:

Principle 1: Visualize the Constraint

Students cannot manage what they cannot see. Mela provides a clear, at-a-glance dashboard showing the monthly stipend, budgeted essential expenses, and the remaining buffer. A calendar projection maps out the user's expected balance day-by-day, highlighting potential low points before they occur. When Biruk can see that his balance will drop below 100 Birr on the 25th, he can adjust his spending earlier rather than scrambling at the last minute.

Principle 2: Prioritize Necessities First

In a world of scarcity, prioritization is survival. Mela helps students identify and protect essential spending categories; food, academic materials, personal care, and transportation before anything else. Users define estimated monthly costs for these core categories during setup, and the app tracks spending against them throughout the month. For Hiwot, who is learning to budget for the first time, this structured approach provides a safety net that prevents costly mistakes.

Principle 3: Provide Proactive, Non-Judgmental Guidance

Mela does not shame users for their spending; it guides them toward better decisions with timely, helpful alerts. When a user approaches their budget limit in any category, the app sends a neutral notification: "You've used 75% of your food budget. 10 days remaining." This allows students to adjust their pace without guilt or shame—like a trusted friend offering a gentle reminder.

Principle 4: Enable Micro-Saving as a Habit

Saving money feels impossible when every Birr is already allocated to necessities. Mela challenges this by focusing on micro-saving; small, consistent, achievable goals that build over time. A user can set a goal like "Save 300 Birr for graduation photocopying in 2 months," and the app calculates the tiny daily contribution required: "Save just 5 Birr per day." This approach makes saving feel possible and builds a meaningful financial buffer over time.

Principle 5: Adapt to the Student Lifecycle

A first-year student and a final-year student have different financial needs. Mela adapts accordingly. For first-year students like Hiwot, the focus is on education and habit formation; simple tracking and basic budgeting. For upper-year students like Biruk, the app offers advanced features like project-specific expense tracking and goal setting for graduation costs. This ensures Mela remains relevant from the first month of university to the final semester.

Bringing It All Together

By combining these five principles, Mela transforms financial management from a source of chronic stress into a source of empowerment. Students no longer have to rely on mental math, scattered notes, or generic apps that do not understand their reality. Instead, they have a dedicated tool that supports their goals and helps them navigate the financial challenges of university life with clarity and confidence. The ultimate goal is simple: to free students' mental energy from the constant burden of financial worry so they can focus on what truly matters—their education, their growth, and their future.

This version maintains the expanded core principles while being more streamlined and focused, making it suitable for a proposal document that balances depth with conciseness.

5. Core Features (Minimum Viable Product - MVP)

To deliver a useful and focused first version, we will concentrate on these 5 essential features, each directly addressing the core problem:

- **1. Stipend & Essentials Setup:**

    - **User Authentication:** Simple sign-up using email or phone number.
    - **Financial Profile:** Upon first login, the user inputs their **monthly stipend amount**.
    - **Essentials Budgeting:** The user then defines their estimated monthly costs for core necessity categories:
        - *Daily Food (Cafeteria/Main Meals)*
        - *Academic Materials (Printing, Stationery)*
        - *Personal Care (Soap, Sanitary Products)*
        - *Transportation (Local)*
    - This setup establishes the user's financial baseline and constraints.
- **2. Intelligent Dashboard & Projection:**
    - A clear, at-a-glance dashboard showing:
        - **Stipend Received:** The total for the month.
        - **Essential Expenses Budgeted:** The sum of the user's defined essential categories.
        - **Remaining Buffer:** The amount left after accounting for essentials.
    - A **calendar view** that projects the user's balance day-by-day, factoring in their stipend and their planned spending schedule, helping them see potential low points before they happen.
- **3. Real-Time Expense Tracker:**
    - Quick and simple logging of daily expenses.

    - Expenses are tagged with the pre-defined essential categories (Food, Academic, Personal Care, Transport). This reinforces prioritization.
    - A visual chart shows the month-to-date spending breakdown by category, allowing students to see exactly where their stipend is going.
- **4. Adaptive Spending Alerts:**
    - The app will send proactive, non-judgmental alerts to prevent overspending.
    - **Example Alerts:**
        - "You've used 75% of your 'Daily Food' budget for this month. 10 days remaining."
        - "Your current spending rate on 'Transport' is higher than planned. Consider reviewing your trips this week."
    - This feature provides real-time feedback, helping students self-correct before it's too late.
- **5. Micro-Saving Goals:**
    - A dedicated feature to set small, achievable savings goals. The emphasis is on *micro*-savings, acknowledging the tight budget.
    - A user could set a goal like "Save 300 Birr for Graduation Photocopying in 2 months."
    - The app calculates a tiny, daily or weekly saving amount (e.g., "Save just 5 Birr a day") and shows progress. This makes saving feel possible and builds a crucial financial buffer over time.

6. Team Members & "Everyone Codes" Roles

Our team consists of 3 members. To adhere to the "Everyone Codes" principle, each member takes ownership of a primary domain while actively contributing to all areas of development. This ensures shared knowledge, collective responsibility, and a robust final product.

Full Name	Primary Technical Role	Core Responsibilities	Cross-Functional Contributions
Lediya Ababu	UI/UX & Frontend Lead	- Translating Figma wireframes into functional UI using Jetpack Compose. - Implementing all user-facing screens and navigation flows. - Ensuring a smooth, intuitive, and responsive user experience.	- Assisting with local data persistence logic. - Contributing to user input validation and form handling. - Participating in testing of UI components and user flows.
Hana Daniel	Logic & Data Layer Lead	- Designing the Room database schema for users, expenses, budgets, and goals. - Implementing all business logic: projection algorithms, spending calculations, alert triggers, and savings goal tracking. - Managing the app's state and ensuring data consistency.	- Collaborating on UI implementation for data-heavy screens. - Assisting with database query optimization. - Contributing to unit testing for core calculations.
Halleluya Belay	Infrastructure & Integration Lead	- Setting up and managing the GitHub repository (branching strategy, .gitignore, collaboration). - Implementing user authentication (email/phone) and any secure data handling. - Managing project documentation and ensuring Jira tasks are up to date. - Leading testing efforts (unit, integration, UI) and coordinating the final build.	- Contributing to UI implementation for authentication flows. - Assisting with integration between authentication and local database. - Writing and maintaining test suites.

Collaboration Approach:

- All three members will participate in daily stand-ups and sprint planning.
- Code reviews will be conducted for every pull request, ensuring all team members are familiar with all parts of the codebase.
- Knowledge sharing sessions will be held to cover key implementations (e.g., Team Member 2 will walk the team through the projection algorithm once implemented).
- This structure ensures that if one member is unavailable, the others can confidently contribute to any area of the project.